On AI and Organisations

Exploring Human-AI Collaboration and Systems Design

On AI and Organisations

Over the past few days I've been reading *The Fifth Discipline* by Peter Senge, and one idea has stayed with me: **an invention and an innovation** are not the same thing.

An invention can exist for years before it becomes an innovation. It only becomes an innovation when it can be replicated at meaningful scale and practical cost .Thus, changing the way people and organisations operate.

That made me think about AI.

Right now, a lot of organisations are racing to "adopt AI",but I keep wondering whether we're moving faster than our understanding of the systems we're trying to redesign.

We tell people to "learn AI." We deploy copilots. We build agents.

But are we redesigning the human workflows around them or simply inserting AI into systems that were never designed for this way of working?

Maybe the real challenge isn't AI implementation. Maybe it's **systems design**.

How do we redesign human systems so that AI expands human capability rather than simply automating existing work?

From Invention to Innovation: Designing Human Systems for AI

For much of history, we inherited systems that shaped how we learned, worked, governed, and collaborated.

Changing those systems was often slow, expensive, and reserved for a few.

Today, AI gives us a new capability: the ability to rapidly imagine, prototype, and test alternatives.

The opportunity isn't simply to build more intelligent machines, but to become more intentional designers of the human systems those machines will inhabit.

The question is no longer whether the world can change. The question is what kind of world we choose to build together.

Human-AI Collaboration: Designing for Better Judgment

Artificial intelligence is changing how we work.

But I don't think the biggest question is whether AI will become more capable. I think the bigger question is: **how do we design human systems that become better because AI is part of them?**

For decades we've measured technological progress by efficiency, automation, and productivity. Those metrics matter but they don't tell the whole story.

As AI becomes embedded in research, education, healthcare, business, and government, we need to ask different questions.

Does the system strengthen or weaken **human judgment**? Does it develop expertise or gradually **deskills** it? Do workflows encourage **critical thinking** rather than passive acceptance?

Technology never exists in isolation. It becomes part of human systems, and those systems shape how people think, collaborate, and make decisions.

Maybe the future of AI won't be defined solely by what these systems become capable of doing. Maybe it will be defined by how intentionally we design the human systems within which they operate.

AILIRA: AI Literature Review Associate

Over the past week, I've been reading foundational papers on human–AI collaboration while developing a research proposal.

Very quickly, I realised something: the challenge wasn't simply reading papers. It was the **process surrounding the reading**.

Finding the right literature. Understanding dense academic language. Comparing arguments across multiple papers. Taking structured notes. Tracking ideas. Knowing what deserves a deep read and what can be reviewed more efficiently.

So instead of asking, "Can AI write my literature review?" I asked a different question: **How can AI help me become a better researcher without replacing the thinking?**

That question led me to build the first version of AILIRA (AI Literature Review Associate).

Importantly, AILIRA isn't designed to write literature reviews. Its purpose is to support everything that comes **before** writing:

- Search and import relevant literature
- Generate a structured scholarly analysis of each paper
- Explain complex concepts in plain language
- Discuss papers interactively through an AI research associate
- Capture research notes and reflections
- Compare multiple papers through literature synthesis

After using it on two foundational papers, one observation surprised me. The biggest benefit wasn't speed it was reducing the **cognitive effort** required to begin reading.

Instead of opening a paper with little orientation, I entered it already understanding its methodology, structure, and key ideas. That let me spend more time thinking critically and connecting ideas, rather than decoding dense academic writing.

For me, that's a more interesting direction for AI: not replacing scholarship, but creating systems that help researchers think more deeply, ask better questions, and engage more meaningfully with the literature.

This is only Version 1, built through real use over the last few days, and I'm excited to keep refining it as part of my broader research into human–AI collaboration.

Redesigning Processes: The Organogram Builder

A client needed an organogram. The challenge was that not all the information was available yet.

Instead of waiting for missing details (and risking a last-minute rush), I asked a different question: **what if the client could update the information themselves as it became available?**

A few hours later, what started as a simple organogram request became an interactive Organogram Builder with editable structures, export functionality, and full client control.

One lesson I've learned is that sometimes the best solution isn't to work harder within a process. It's to **redesign the process** itself.

This project is one of several digital artefacts I've been building as I explore how technology can turn information into usable tools.

Still refining. Still learning. Still building.

Aether: Building Multi-Participant Voice AI

Most conversational AI systems assume a single active speaker, making natural multi-participant interaction difficult. Aether explores a different architectural approach by introducing an explicit conversation state layer that manages participant identity, conversational continuity, and turn-taking independently of the language model.

Rather than relying solely on speaker diarization or prompting an LLM to infer identity from transcripts, Aether investigates how stateful orchestration can improve multi-user conversational experiences.

The project serves as an exploration of AI systems architecture, focusing on conversation management, participant state, and human–AI interaction rather than model capability alone.

A Nod to Kimi (Moonshot AI)

A nod of appreciation to Kimi (Moonshot AI) and the team behind it.

One thing that stood out to me is how accessible they've made AI agents feel. For many people, agents are still something they hear about in demos, videos, and technical discussions.

Kimi let me actually experience what an agent can do: I could describe what I wanted, watch it begin creating the product, and then refine and iterate on the result before hitting the usage limit.

The experience wasn't perfect, but it was one of the first times I felt the gap between an idea and a working artefact shrink so dramatically.

What impressed me most wasn't necessarily the final output. It was how quickly I could move from **intention** to **creation**.

Making agent capabilities accessible to more people is an important step forward, and Moonshot deserves credit for that.

AI-Assisted Creative Work: Interior Design Portfolio

I recently completed an interior design portfolio project that combined portfolio design, branding, creative direction, and **AI-assisted visual development**.

The objective was to transform a collection of project photographs into a professional portfolio that could showcase completed work and support future business opportunities.

One of the biggest challenges was that many source images were captured on-site using a phone camera. They documented the spaces, but they weren't always presentation-ready: some images were tightly cropped, key elements were cut off, perspectives were limited, and quality varied between projects.

To address this, I analysed multiple photographs of the same spaces and developed portfolio-ready visualisations that created wider, more cohesive perspectives while staying faithful to the original design intent.

This project included:

- **Portfolio architecture** and layout design
- **Collection pages** and project showcase development
- **AI-assisted image development** and refinement
- **Logo** and brand identity development
- **Visual consistency** and quality control
- **Creative direction** throughout the project

One of the biggest misconceptions about AI-assisted creative work is that the technology does the thinking. In reality, the value comes from understanding the desired outcome, identifying what needs to be preserved, recognising what needs refinement, and guiding the process through iterations until the final result accurately represents the original space and vision.

The images shown here are only a small excerpt of the final portfolio, and they highlight part of the creative and technical process behind the work.

Research Proposal: Designing Human-AI Systems That Strengthen Judgment

As AI becomes embedded in organisations and everyday life, the central question is no longer simply how to build more capable systems, but how to responsibly integrate those systems into human life.

This is fundamentally a **systems design challenge**—because the design choices we make determine how work actually gets done, not just what the technology can do in isolation.

Those choices will shape productivity, but also how people exercise **human judgment**, develop expertise, share responsibility, and decide what (and who) to trust inside the systems they work within.

The proposal argues that the objective of human–AI collaboration should not be the maximisation of automation, but the **maximization of human capability**.

Rather than asking how AI can replace more human work, it asks how human–AI systems can be intentionally designed to preserve and **strengthen human judgment** while still helping people work more effectively alongside increasingly capable AI systems.

In other words, the aim is to design workflows where AI supports reasoning, verification, and reflection—without quietly shifting responsibility away from the human.

Methodologically, the research proposes a qualitative single-case study following an independent researcher conducting an authentic research project while collaborating with multiple AI systems assigned distinct roles.

The collaborative workflow itself becomes the object of investigation, so the **research process becomes empirical evidence** rather than a background activity.

Across the study, the work will document how cognitive tasks are delegated, how trust is calibrated, how AI-generated outputs are verified, and how uncertainty, disagreement, and error are identified and resolved over time.

This perspective is especially important in South Africa and other **resource-constrained contexts** where constraints are already part of everyday organisational reality.

Much global AI discourse assumes stable infrastructure, abundant computational resources, and mature digital ecosystems, but many organisations are integrating AI amid limitations in skills, services, and institutional capacity.

In such settings, simply importing AI systems without redesigning the surrounding human systems risks amplifying existing challenges instead of reducing them.

Ultimately, the proposal returns to a broader claim: the future of AI will not be determined only by capability, but by **how intentionally we choose to design the human systems** within which it operates.

Responsible innovation should therefore be evaluated by whether it strengthens human judgment, preserves human agency, and helps people become better thinkers and decision-makers—not just faster producers of outputs.

Research: Understanding Drift in Extended AI Conversations

Over the past several months, I've been studying a recurring phenomenon I observed across sustained conversations with Claude, Gemini, DeepSeek, and ChatGPT.

I refer to this phenomenon as **drift**: a gradual shift in an LLM's reasoning or conversational trajectory over time during extended, high-context dialogue.

What became especially interesting is that drift didn't show up as a single failure mode. In transcripts, it tended to lead in two directions:

- **Generative drift** — novel synthesis, unexpected connections, co-creative ideation

- **Destabilising drift** — hallucination, repetition, narrative fixation, or reduced grounding

This working paper proposes drift as an interactional phenomenon emerging through sustained human–AI dialogue, and explores what it might mean for long-form reasoning, AI collaboration, and future research into human–LLM interaction.

It’s an exploratory framework—for now—but it’s helping me name something I think many of us have felt in extended use: the system doesn’t just answer; it **moves** over time, and that movement can be either useful or risky depending on how the surrounding workflow is designed.

The Microzoo Error: Detecting Communication Breakdowns

Yesterday I stumbled across a small error that turned into a surprisingly useful observation about human–AI collaboration.

I was using voice mode while discussing research. I intended to say “micro-zoom.” The speech transcription recorded it as “microzoo.” I didn’t notice the mistake before sending the message.

The AI accepted “microzoo” as the input and then built a coherent line of reasoning around it, even developing a zoo-based analogy that fit the conversation surprisingly well.

Only later did I notice the transcription error. Once corrected, the entire interpretation changed.

What struck me wasn’t that the AI made a mistake. It reasoned coherently from the information it received. The real question is this: **how should human–AI systems detect when a plausible interpretation might still be the wrong interpretation?**

In human conversation, we constantly repair misunderstandings. We ask “Did you mean...?” We use hesitation, tone of voice, facial expressions, and shared context to course-correct in real time.

Many AI systems get only the processed transcript. If that transcript is wrong, the reasoning that follows may be perfectly logical but built on an incorrect foundation.

As AI becomes more deeply integrated into organisations and knowledge work, I suspect one of the biggest design challenges isn’t just improving reasoning, but improving the ability to **detect and repair communication breakdowns** before they propagate through a workflow.

It was a tiny transcription error. But it raised a much bigger question about communication, context, and the future of human–AI collaboration.

Conclusion

These explorations represent an ongoing inquiry into how we design human systems in the age of AI.

The question isn’t just what AI can do. It’s how we intentionally shape the systems within which AI operates. From research tools to organisational design, from communication breakdowns to creative collaboration, each project asks: **How do we expand human capability rather than simply automate existing work?**